Seaglass Globe™

Military Symbology on a Real Coordinate System, Behind a Loopback-Only Boundary

For programs where a picture of the Earth is not enough: MIL-STD-2525C symbols and tactical graphics drawn on a WGS-84 ellipsoid held in double precision, scenarios you can step through on a timeline, measurements that name their algorithm, coordinates read as DD, DMS, UTM and MGRS, and the Forge suite's GPU compute drawn where you are looking. It runs on one Windows x64 workstation with a real GPU. Its API, like the agent's, binds to 127.0.0.1 behind a per-install bearer token; neither is exposed to the LAN.

2525C

symbols and tactical graphics, in the box

the symbol set installs with the Globe; 2525D and E codes are translated to C for drawing, and every loss in the translation is reported

MGRS

and UTM, DMS, DD — from one cursor

the military grid reference read straight off the status bar, checked to 1 cm against PROJ and GEOTRANS; the readout re-forms as you switch

127.0.0.1

the whole boundary is local

the Globe's API and the agent's bind to loopback behind per-install bearer tokens, and neither is exposed to the LAN

Symbology you can read

Units draw with their frames and text amplifiers, headquarters, task-force and feint-dummy decorations, in Standard, Monochrome or Night colour. Operational condition is painted into the symbol — muted when damaged, ashen when destroyed — and a unit in dead ground is ghosted, not dropped. Phase lines, objectives, axes of advance and range fans draw as tactical graphics, anchored to terrain or to another unit, with handles you can drag and undo.

A coordinate system an evaluator can check

The datum is not a label: geodetic ↔ ECEF runs in double precision, and altitude is ellipsoidal, along the geodetic normal. Two clicks return distance, forward and back azimuth on the ellipsoid by Karney's geodesic method, and the badge names it; close a polygon and the area is Karney's too. Save a measurement and the GeoJSON carries the algorithm beside the geometry, so a reviewer can check it with their own tools.

Runs where the work is

One workstation, everything local. The Seaglass Control Center™ installs the Globe beside the engines, with nothing to install first, and keeps keys in the Windows Credential Manager, never on disk. Visited regions serve from a tile cache that never expires, any WMS, WMTS, XYZ or MVT service you host is added by URL, and your own elevation enters through the ForgeData™ catalog.

Checked, not asserted — release 1.2.6. A testing and fixes release, held to outside references rather than to its own code. Every measurement now runs on the ellipsoid, geodesics by Karney's method; the MGRS readout truncates as MGRS does, and 49 UTM and MGRS points are checked against PROJ and GEOTRANS to a centimetre. A DEM in US survey feet reads in its own units instead of 3.28 times too high. A click on a draped feature on high ground resolves on the terrain, not the ellipsoid behind it, and a line across the antimeridian is centred on itself rather than on the far side of the Earth.

CONTINUED ON REVERSE →

WHAT PROGRAMS GET

Exercises on a timeline

A scenario is a timeline: play it in real time or compressed, forward or in reverse, or jump from event to event while units move along their tracks. The suite includes a brigade-advance demonstration with notional positions, opened from File → Open scenario. ForgeRelay™, a separate opt-in component, records live DIS simulation traffic as ground truth and a Cursor-on-Target feed as one side's picture, in the same scenario format; the Globe is pointed at a recording by hand. Any scenario exports to a GeoPackage QGIS can open.

Terrain work on real elevation, re-run while you hold the slider

Terrain does not stop at the waterline: the depth tint below it reads the same elevation data as the mountains above. Point at the ground and ask what elevation data exists there; it ranks what it can reach and hands you a document with that file set as its source, named, with its resolution. Type a contour interval; drag the sun's altitude and the hillshade re-computes and re-drapes while you hold it.

Your own data, with its attributes

GeoJSON, KML and shapefiles open on a layer stack that is yours — a zipped Shapefile is read in memory, so a hostile archive cannot write a file — and GeoPackage opens through the engine. Each file can be styled — stroke, fill, point size, a label field — with a legend in the attribution strip, and an attribute table docks under the map: find across every cell, sort, click a row to frame and ring the feature, export as CSV or GeoJSON.

DEPLOYMENT POSTURE

Local

the MCP API and the record-ingest route share one loopback port behind a per-install bearer; a missing token gets an empty 401, and a non-loopback Host or a cross-site page is refused, token or not

Contained

the compute engines run as child processes inside ForgeGIS's read and write path sandbox; a missing engine degrades the compute tools, never the globe

Bounded

every tool call is time-limited and panic-guarded, and every engine call has a deadline: a wedged engine returns an error to the caller, and the globe window stays up

Footprint. Windows x64 with a real GPU — Vulkan or DX12, whichever wgpu picks; an RDP or remote-desktop session usually can't drive it, and a minimised Globe stops drawing. The suite recommends 8 GB of RAM with the Globe installed, because the Globe runs its own copy of the GIS engines, plus a few gigabytes of disk for a tile cache that grows with where you have flown. Installed and started by the Seaglass Control Center on port 7900, loopback only.

Honest by construction. No operator, dataset or tool counts — they vary with what is installed. The symbol art is 2525C; D and E codes are drawn through it. The historical battle datasets used in development do not ship. A large drawn result from a projected source is declined rather than drawn wrongly. The flat projections are spherical; the ellipsoid lives in the geodesy and the measure tool. The live feeds and satellite catalogue shown in the release film are showcase builds, not part of the installed suite, and every twin run is a model.

Request the Technical Brief and ask about evaluation access

The Technical Brief covers symbology and scenarios, the geodesy and precision model, rendering, the data and feed surface, the MCP agent surface, the compute and twin bridges, security posture and deployment. Seaglass Globe is an opt-in component of the Forge suite, installed by the Seaglass Control Center, and is not sold separately. The Forge suite is commercially licensed; evaluation builds are available to qualified teams.

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